

# 12 Learning Disabilities

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## Introduction

Learning disabilities are neurobiologically based developmental disorders that affect the brain's ability to receive, process, store, and respond to information. They typically arise in childhood, persist into adulthood, and are associated with unexpected academic underachievement. This underachievement is considered "unexpected" when it is not explained by low intelligence, sensory impairments, emotional disturbances, or lack of opportunities to learn. Learning disabilities occur along a continuum, with variability in severity and characteristic features, rather than as discrete, dichotomous entities.

This chapter focuses on three learning disabilities that have been broken into several subtypes:

- **Reading disabilities** (word recognition/dyslexia, reading fluency, comprehension)
- **Mathematics disabilities** (computation, problem solving)
- **Written expression disabilities** (handwriting, spelling, composition)

Some neuropsychologists have argued that nonverbal learning disabilities form a coherent subtype that can be distinguished from other disorders with similar impact, such as mathematics disabilities or attention-deficit/hyperactivity disorder (ADHD; see also Chapter 13, Attention-Deficit/Hyperactivity Disorder); however, recent critical reviews have concluded that there is insufficient empirical and clinical support for such a distinction. The prevailing (narrower) definition places the emphasis on learning disabilities that cause academic dysfunction. For these reasons and due to space limitations, nonverbal learning disabilities are not covered here.

## Definitions

The most recent legal definition of learning disabilities was included in the Individuals with Disabilities Education Improvement Act of 2004. It states:

Specific learning disability means a disorder in one or more of the basic psychological processes involved in understanding or in using language, spoken or written, that may manifest itself in an imperfect ability to listen, think, speak, read, write, spell, or do mathematical calculations. The term includes such conditions as perceptual handicaps, brain injury, minimal brain dysfunction, dyslexia, and developmental aphasia. The term does not include children who have learning problems that are primarily the result of visual, hearing, or motor handicaps; of mental retardation; of emotional disturbance; or of environmental, cultural, or economic disadvantage.

The Rehabilitation Services Administration has incorporated a different definition of learning disabilities specific to adults into disability law:

A specific learning disability is a disorder in one or more of the central nervous system processes involved in perceiving, understanding, and/or using concepts through verbal (spoken

or written) language or nonverbal means. This disorder manifests itself with a deficit in one or more of the following areas: attention, reasoning, processing, memory, communication, reading, writing, spelling, calculation, coordination, social competence, and emotional maturity.

## DSM-5 Definition of Specific Learning Disorder

The new diagnostic criteria in the American Psychological Association's *Diagnostic and Statistical Manual of Mental Disorders* (DSM-5) for Specific Learning Disorder reflect a hybrid model of identification. They also reflect recognition that individuals may "grow into" their learning deficits and so problems may not be fully manifested until a later age.

- Diagnosis is made using a synthesis of the individual's history (development, medical, family, education), psychoeducational reports of test scores and observations, and *response to intervention*. A Specific Learning Disorder is now defined as:
  - Difficulties learning and using academic skills, as indicated by the presence of at least one of the following symptoms that have persisted for at least 6 months, despite the provision of interventions that target those difficulties.
    - Inaccurate or slow and effortful word reading (e.g., reads single words aloud incorrectly or slowly and hesitantly, frequently guesses words, has difficulty sounding out words).
    - Difficulty understanding the meaning of what is read (e.g., may read text accurately but not understand the sequence, relationships, inferences, or deeper meanings of what is read).
    - Difficulties with spelling (e.g., may add, omit, or substitute vowels or consonants).
    - Difficulties with written expression (e.g., makes multiple grammatical or punctuation errors within sentences, employs poor paragraph organization; written expression of ideas lacks clarity).
    - Difficulties mastering number sense, number facts, or calculation (e.g., has poor understanding of numbers, their magnitude, and relationships; counts on fingers to add single-digit numbers instead of recalling the math fact as peers do; gets lost in the midst of arithmetic computation and may switch procedures).
    - Difficulty with mathematical reasoning (e.g., has severe difficulty applying mathematical concepts, facts, or procedures to solve quantitative problems).
- The affected academic skills are substantially and quantifiably below those expected for the individual's chronological age, and cause significant interference with academic or occupational performance, or with activities of daily living, as confirmed by individually administered standardized achievement measures and comprehensive clinical assessment. For individuals age 17 years and older, a documented history of impairing learning difficulties may be substituted for the standardized assessment.
- The learning difficulties begin during the school-age years but may not become fully manifested until the demands for those affected academic skills exceed the individual's limited capacities (e.g., as in timed tests, reading or writing lengthy complex reports for a tight deadline, excessively heavy academic loads).
- The learning difficulties are not better accounted for by intellectual disabilities (see also Chapter 11, Intellectual Disability), uncorrected visual or auditory acuity, other mental or neurological disorders, psychosocial adversity, lack of proficiency in the language of academic instruction, or inadequate educational instruction.

## Incidence Rates

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Approximately 15–20% of the US population (i.e., more than one in seven Americans) has a learning disability. The number of affected individuals varies depending on the assessment methods and the threshold used to define the “deficit” in academic achievement.

- Difficulties with basic reading and language skills make up the most common learning disabilities. As many as 80% of individuals with learning disabilities have marked reading problems.
- Males are about 1.5 times as likely as females to have reading disabilities.
- Learning disabilities often run in families, with heritability for reading ranging from approximately 0.3 to 0.9, and for math ranging from 0.5 to 0.8.

## Presentation, Disorder Course, and Recovery

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- For most individuals with learning disabilities, the deficits first have an impact in childhood, usually in the preschool or early elementary school years.
- A learning disability typically represents a persistent functional deficit, rather than a developmental lag. These disabilities frequently persist over time, despite intervention efforts, and typically do not spontaneously remit with time or age.
  - For some, the manifestations and impact may not become evident until later in childhood, in the teenage years, or even in the adult years.
  - In those diagnosed for the first time as adults, the disorders tend to be milder, rather than reaching the level of a full-fledged disability that requires significant accommodations or supports.
- Among individuals with early-onset learning disabilities who have received intervention by early elementary school, deficits in word reading accuracy and/or mathematics calculation can improve; however, deficits in phonological processing, automaticity of word recognition, and reading fluency tend to persist.

## Diagnosis of Learning Disabilities

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### Aptitude–Achievement Discrepancy Model

- Many states and the DSM-IV identify children with learning disabilities based on a discrepancy between measured intelligence and a specific academic skill.
- The discrepancy model has been challenged from a research perspective. In particular, individuals diagnosed using a discrepancy method do not differ functionally or behaviorally from individuals diagnosed using a low-achievement method alone.

### Low-Achievement Model

- Models based primarily on low achievement have been shown to have substantial validity and are frequently used in research.
- Because youth who perform below average on achievement tests (and do not have intellectual disability) can be differentiated on external variables, low achievement is considered a viable definition for learning disability.

## Intraindividual Differences Model

- This model takes into account individual profiles of strengths and weaknesses in cognitive functioning and information processing. Students are identified as having learning disabilities when their profiles are markedly uneven.
- The validity of this model has been challenged.
  - Cognitive profiles derived from neuropsychological assessment do not translate readily into individualized instruction.
  - It assumes that a “flat” profile is inconsistent with a learning disability.
  - Variability (“test scatter”) is the rule rather than the exception and, alone, cannot be used to diagnose a learning disability.

## Response to Intervention (RTI) Model

- The premise of RTI is that students should receive adequate instruction with appropriate monitoring before being considered to have a learning disability.
- In the RTI model, the response of all students to standard and adequate instruction is evaluated from the start of school.
  - Those who do not acquire a specific academic skill undergo selective assessment of that skill (often within the context of curriculum-based measurement).
  - Assessment can include limited neuropsychological testing of the abilities important for that academic skill (e.g., phonological awareness for reading). Following assessment, a new instructional plan is developed and implemented, its impact is assessed, and the instructional plan is further modified as needed.
- A potential problem of the RTI model is that little consensus exists regarding how to define “inadequate response to appropriate instruction.” There also is no research support for an RTI approach in adults.

# Section 1: Reading Disabilities

## Definition

The scientific literature has shifted to definitions that reflect more specific types of reading problems. Three subtypes are reflected in the descriptors for reading disabilities included in diagnostic criteria in the DSM-5: word decoding (dyslexia), reading fluency, and reading comprehension.

- The most common reading disability, and the only one for which there is an evidence-based definition, is dyslexia.

Dyslexia is a specific learning disability that is neurobiological in origin. It is characterized by difficulties with accurate and/or fluent word recognition and by poor spelling and decoding abilities. These difficulties typically result from a deficit in the phonological component of language that is often unexpected in relation to other cognitive abilities and the provision of effective classroom instruction. Secondary consequences may include problems in reading comprehension and reduced reading experience that can impede growth of vocabulary and background knowledge.

## DSM-5 Descriptive Feature Specifiers for Impairment in Reading

The following descriptive specifiers highlight the domains of academic difficulties and their subskills that are impaired at the time of assessment:

- Word reading accuracy
- Reading rate or fluency
- Reading comprehension

## Neuropathology

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- A number of candidate genes have been linked to dyslexia. Some of the most promising genes have been implicated in abnormalities in neuronal migration.
- Children at genetic risk for development of dyslexia display differences (as early as a few days to 6 months of age) in brain response and orienting to speech sounds.
- Postmortem studies of individuals with developmental dyslexia revealed consistent (unexpected) symmetry of the planum temporale, polymicrogyria of the left planum temporale, and cortical dysplasias in the left hemisphere.
- Some researchers have implicated a region in the left occipitotemporal cortex, including the middle part of the left fusiform gyrus, devoted to rapid processing of words, referred to as the *visual word form area*.
- Recent functional neuroimaging studies have also shown a profile of left-hemisphere abnormalities in children and adults with dyslexia.
  - During tasks requiring phonological analysis, underactivation is seen in Wernicke's area, angular gyrus, and striate cortex, with concurrent overactivation in the inferior frontal gyrus—sometimes referred to as the *neural signature of dyslexia*.
  - During word processing, reduced activity occurs in the left occipitotemporal junction.
  - Children with poor reading fluency have less activation in left frontal regions than do children with adequate reading fluency.

## Incidence Rates

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Reading disabilities are estimated to occur in 10–15% of school-aged children and adolescents; however, estimates differ depending on the methods used for defining impairment.

## Comorbid Conditions with Dyslexia

- Approximately 55% of children with dyslexia have impaired oral language skills, and 51% of children with language disorders exhibit deficient reading.
- Deficits in language skills are also common in adults with dyslexia.
- Incidence of mathematics disability in those with dyslexia is as high as 56%.
- Approximately 25–40% of children with dyslexia have ADHD, and approximately 25–40% of children with ADHD have dyslexia. The two disorders co-occur more often than would be expected by chance.

- Deficits in processing speed have been reported in both dyslexia and ADHD and are independent of specific deficits in phonological awareness and language skills, perhaps a cognitive reflection of the genetic overlap.

## Presentation and Disorder Course

### Skills Considered Fundamental for Acquisition of Reading

- **Phonological Awareness** The individual's ability to understand that words are comprised of specific speech sounds (phonemes). Phonological skills involve the ability to parse out the component sounds of words and to put them back together. Most individuals with dyslexia have deficient phonological awareness and have difficulty converting graphemes (written units of sound) into phonemes.
- **Decoding** The individual converts letters into sounds and combines them to form a recognizable word. The ability to convert letters to sounds is based on knowledge of the alphabet and phonological awareness.
- **Sight Reading** The individual retrieves or recognizes words from sight that are already known. As this skill develops, the recognition becomes faster and more automatic, and the number of sight words grows.
- **Prediction** The reader applies context, linguistic and background knowledge, and memory to text to anticipate or guess the meaning of unknown words. This is a top-down (executive) skill, facilitated by knowledge of word-letter associations.

### Automaticity, Rapid Naming, and Reading

- Rapid automatized naming (RAN) has been shown to be strongly predictive of reading, independent of phonological awareness.
  - Failure to automatize skills necessary for rapid serial naming is considered a core difficulty among children with reading disabilities.
  - The most widely used measures of RAN require the individual to quickly and accurately name sets of overlearned visual stimuli (e.g., letters, simple objects, digits, and colors). For adults, however, measures of object naming are likely to be most sensitive, because naming of letters, digits, and colors is far more automatic.

### The Double-Deficit Model of Reading Disabilities

- The model argues for the presence of two "single-deficit" subtypes of reading disabilities with more limited reading impairments and one "double-deficit" subtype with more pervasive and severe impairments.
- Rapid naming speed (single-deficit) and phonological awareness (single-deficit) variables contribute uniquely to different aspects of reading according to this model.
  - The implications of processing speed as a core deficit are also considered.
- There is some limited support for this model in adults.

## Reading Fluency

- Development of reading fluency is critical for becoming a competent reader.
- Reading fluency depends on skills that include phonological awareness, phonics, automaticity of sound–symbol relationship retrieval, decoding, recognition of sight words, vocabulary, and processing speed.
- Individuals with dyslexia typically have problems with fluent reading of text; however, poor reading fluency is also observed in individuals who do not have basic word recognition difficulties (e.g., children with ADHD commonly have poor reading fluency even in the absence of dyslexia).
- For some, fluency may normalize, especially if the problem is addressed by third grade, but for others, reading and understanding text remains difficult throughout school and into adulthood.
- Deficits in reading comprehension tend to improve among individuals who have had direct instruction in methods to develop vocabulary, reading fluency, and strategy use.
- The primary problem for adults is slow, inefficient reading, even when decoding and single-word reading are adequate. This can be due to associated problems with spoken language and working memory.

## Reading Comprehension

- According to the 2011 National Assessment of Educational Progress (NAEP), 34% of fourth-graders and 24% of eighth-graders performed at the “below basic” level in reading comprehension.
- Deficits related to reading comprehension tend to fit into three categories:
  - Basic word recognition skills
  - Broad oral language skills
  - Executive functions (including working memory, planning, organized memory search, estimation)
- The “simple view” of reading emphasizes the necessity of both word-level and other language skills for proficient reading comprehension.
- Reading comprehension is dependent on orthographic (visual word form) and morphologic (word meaning) processing, in addition to phonologic processing.
- Deficits in attention and executive functioning in children and adults with reading disorders can prevent smooth coordination of the component skills needed for effective reading comprehension.
  - Children with specific reading comprehension disorders (i.e., those with poor comprehension despite intact word recognition skills) have deficits in the planning elements of executive function.
  - Adults can have more difficulty with interpreting what they have read than with literal or fact based understanding.

## Late-Emerging Reading Disabilities

- Learners can struggle with fluency and comprehension in the absence of basic word recognition/decoding problems.
- Approximately 41% of all students with reading disabilities have *late-emerging reading disabilities*; that is, deficits are not evident until at least third grade.
  - This pattern is sometimes known as the “fourth-grade slump,” can be associated with the transition from “learning to read” to “reading to learn,” and may also be related to reduced vocabulary development in students of low socioeconomic status backgrounds.

- *Late-emerging reading disabilities* are often associated with coexisting conditions, especially ADHD.
- Children who received early intervention and showed improvement may start to struggle again with the increased demands and volume of middle and high school reading and when they are expected to work more independently.

## Expectations for Neuropsychological Assessment Results

Although some researchers have concluded that only brief assessment is necessary for any learning disability diagnosis prior to starting treatment, others have concluded that comprehensive assessment is valuable. This is even more important for adults, for whom there can be a variety of reasons for problems with learning. Comprehensive assessment can help:

- Diagnose comorbid conditions
- Determine if a broader intellectual disability better accounts for underachievement
- Determine if another disorder, such as ADHD, better accounts for underachievement
- Determine if emotional factors explain or exacerbate underachievement
- Provide additional targets and recommendations for intervention (e.g., language therapy impairments in spoken language; occupational therapy to improve motor skills, handwriting, and keyboarding; psychotherapy to address emotional issues)

Regarding reading disabilities, Table 12.1 includes the skills most likely to be impaired. By focusing on the expected areas of impairment, assessment can be streamlined.

**TABLE 12.1** Areas of impairment in reading disabilities

- Phonological awareness
- Listening comprehension/immediate memory (single word, discourse)
- General knowledge and vocabulary
- Auditory verbal span
- Executive function (planning, estimation) as related to reading comprehension
- Processing speed
- Word retrieval (confrontation naming, rapid automatized visual naming, timed word generation/word fluency)
- Phonological and orthographic skills (decoding, single-word reading, encoding when reading, spelling)
- Fluency (noncontextual—single words; contextual—text)
- Timed versus extended time reading comprehension (for adolescents and adults)

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## Considerations when Treating Patients with Reading Disabilities

The National Reading Panel reviewed the scientific literature on reading instruction and specifically recommended phonological awareness, phonics, sight word acquisition, vocabulary, and comprehension

of text as essential domains for instruction in normal reading development for all students. They also concluded:

- Oral reading enhances reading fluency better than silent reading practice.
- Instruction in vocabulary should be taught directly, as well as augmented via computer interface.
- Comprehension is developed by a combination of direct instructional methods to develop vocabulary, reading fluency, and strategy use.

## Controversial and/or Unproven Treatments for Reading Disabilities

- ***Vision Therapy and Associated Treatments*** A joint report from the American Academy of Pediatrics and the American Academy of Ophthalmology concluded:

Scientific evidence does not support the claims that visual training, muscle exercises, ocular pursuit-and-tracking exercises, behavioral/perceptual vision therapy, “training” glasses, prisms, and colored (Irlen) lenses and filters are effective direct or indirect treatments for learning disabilities. There is no valid evidence that children who participate in vision therapy are more responsive to educational instruction than children who do not.

- ***Fast ForWord*** This program is a set of computer-based interventions designed to improve oral language and literacy skills in children with language learning weaknesses. The program is based on the principle that language-based learning disabilities are due to a rapid auditory temporal processing deficit that compromises the development of phonological representations. A recent meta-analysis of treatment studies indicated that there was no significant effect of Fast ForWord as a treatment for children’s oral language or reading difficulties.

## Section 2: Mathematics Disabilities

### Definition

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#### DSM-5 Descriptive Feature Specifiers for Impairment in Mathematics

The following descriptive specifiers highlight the domains of academic difficulties and their subskills that are impaired at the time of assessment:

- Number sense
- Memorization of arithmetic facts
- Accurate or fluent calculations
- Accurate math reasoning

#### Other Subtypes of Mathematics Disabilities

- The ***semantic memory subtype*** is characterized by deficient, inaccurate, or inconsistent arithmetic fact retrieval.
- The ***procedural errors subtype*** is characterized by the use of developmentally immature procedures or errors in the execution of procedures.

- The **visuospatial subtype** is characterized by misalignment of numbers or place value errors in decimals. This subtype characterization is based largely on adult lesion data and has not been validated in studies of children.

## Neuropathology

Across a variety of anatomic and functional imaging studies, there is a developmental trend in numerical processing such that young children who are just learning math skills have greater activation in prefrontal regions, whereas older adolescents and adults demonstrate greater functional specialization of the left posterior parietal cortex and decreased dependence on working memory and attentional resources mediated by prefrontal regions.

## Incidence Rates

Emerging data suggest that mathematics disorders are nearly as prevalent as reading disorders and should receive similar attention.

- According to the 2011 NAEP, 18% of fourth-graders and 27% of eighth-graders performed at the “below basic” level in mathematics achievement.

## Comorbid Conditions with Mathematics Disabilities

Reading and mathematics disabilities frequently co-occur.

- Skills in automaticity, processing speed, and working memory needed to read single words are necessary for automatizing math fact retrieval.
- Phonological awareness has been proposed to be associated with mathematics computation in elementary school children.
- In adults, deficits in semantic knowledge, attention, visuospatial skills, and executive functioning affect mathematics performance. Anxiety must also be considered for adults in whom math skills have normalized.

## Genetic and Neurodevelopmental Disorders Associated with Mathematics Disabilities

See also Chapter 15, Chromosomal and Genetic Syndromes, and Chapter 16: Congenital and Acquired Hydrocephalus.

- Turner syndrome
- Fragile X syndrome (girls)
- Congenital hydrocephalus
- Neurofibromatosis type I
- Spina bifida
- Congenital hypothyroidism
- 22q deletion syndromes
- Williams syndrome
- Preterm birth/very low birth weight

# Presentation and Disorder Course

- Many of the same fundamental cognitive processes underlying reading development are also critical for the development of math skills, including language, working memory, automaticity of retrieval of math facts, processing speed, and executive functions (especially planning and estimation).
- Individuals who struggle with math often lack a firm grasp on numerosity (i.e., understanding and recognizing the concept of quantity).
- An important difference between math and reading is that math achievement is cumulative throughout and beyond the elementary school years.

## Expectations for Neuropsychological Assessment Results

Table 12.2 includes the neuropsychological skills most likely to be impaired in mathematics disabilities.

**TABLE 12.2** Areas of impairment in mathematics disabilities

- Attention
- Executive functions: Planning, organization, and problem solving
- Visuospatial skills
- Semantic knowledge
- Fluency for simple math facts (speed of retrieval)
- Written calculation
- Solving word problems (mental, written)
- Mathematical knowledge and concepts

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## Section 3: Written Expression Disabilities

### Definition

A variety of terms have been used to characterize problems with written expression, including developmental output failure, *dysgraphia*, writing disorder, writing problems, disorder of written expression, problems in written expression, writing difficulties, and writing disabilities.

### DSM-5 Descriptive Feature Specifiers for Impairment in Written Expression

The following descriptive specifiers highlight the domains of academic difficulties and their subskills that are impaired at the time of assessment:

- Spelling accuracy
- Grammar and punctuation accuracy
- Clarity or organization of written expression.

# Neuropathology

In children, there are no structural or functional imaging studies that are specific to disorders of written expression.

## Incidence Rates

- Incidence rates of written expression disabilities vary from 7% (using a regression formula) to 15% (using a low-achievement method).
- Boys are two to three more times more likely to have written expression disability than girls.
- Only 25% of those with written expression disabilities do not have dyslexia.

## Comorbid Conditions with Written Expression Disabilities

Most children with learning disabilities in reading or math have significant problems in at least one area of written expression.

## Presentation and Disorder Course

- **Dysgraphia** has been defined as including impairments in legible automatic letter writing, orthographic coding (i.e., storing written words and processing letters in them), and finger sequencing.
- Cognitive phenotypes observed among children with poor written expression include those with only handwriting problems (*dysgraphia*), those with only reading and spelling problems (dyslexia), and those with oral language and written language problems.
  - Students with dyslexia tend to have problems in automatic letter writing and naming that contribute to impaired inhibition and verbal fluency and may explain spelling problems.
- Several components of language are involved in translation of mental information into written format.
  - Transcription involves the subword level (handwriting) and word level (spelling) of written language.
  - Syntax is a level of language that provides structure for organizing words.
- **Processing speed** also plays an important contributing role in written expression disorders. Writing fluency alone may be a better predictor of overall writing success in school than performance on writing samples alone.
- **Working memory** is important in written expression because it allows the individual to engage in multiple cognitive tasks at the same time (e.g., retrieval of information from memory, maintenance of multiple ideas; see also Chapter 5, Domains of Neuropsychological Function and Related Neurobehavioral Disorders).
- Individuals with ADHD can be vulnerable to difficulties with written expression, even in the absence of reading disabilities, as a function of executive function difficulties.
  - Among college students, behaviors associated with ADHD have also been shown to be associated with performance in written expression.
- Factors found to be associated with writing disabilities in adults include slow writing speed, poor handwriting, spelling errors, weak vocabulary, and low verbosity. Men with dyslexia may be more affected in writing skills than women with dyslexia.

# Expectations for Neuropsychological Assessment Results

Table 12.3 includes the neuropsychological skills most likely to be impaired in written expression disabilities.

**TABLE 12.3** Areas of impairment in written expression disabilities

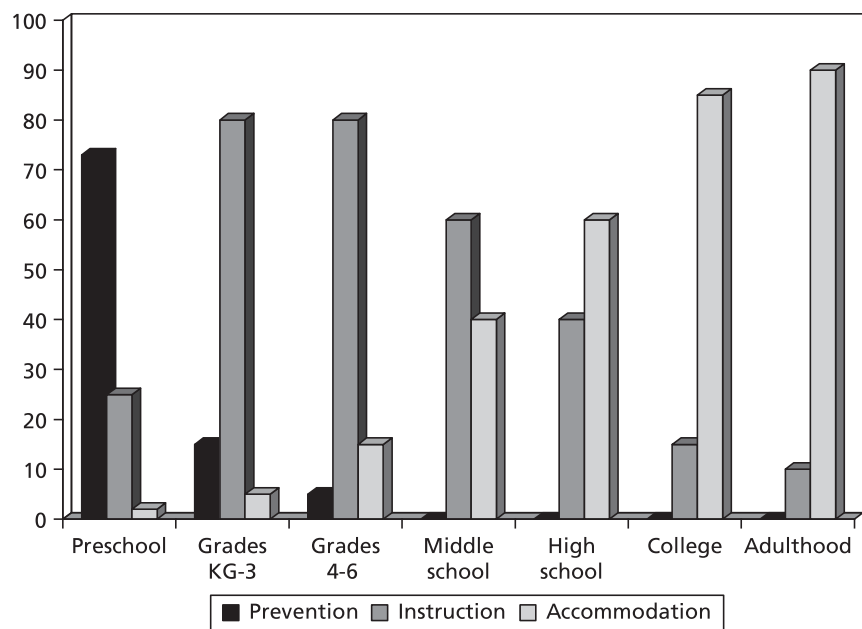
- Motor speed and dexterity
- Handwriting quality (including copying)
- Fluency for writing words and sentences
- Phonological and orthographic skills (encoding, spelling dictation)
- Oral sentence formulation
- Verbal organization (oral, written)
- Essay writing

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## Considerations when Evaluating and Treating Patients with Learning Disabilities

### Shift in Emphasis

As seen in Figure 12.1, when moving from childhood to adolescence to adulthood, the emphasis on how one treats the learning disability changes. For the early years of school, identification and prevention of the learning disability is most important. For children identified later, evidence-based intervention is



**FIGURE 12.1** Prevention, intervention, and accommodation: approaches to learning disabilities over the lifespan. From Morris, R. (2009, February). Discussant for the invited International Neuropsychological Society Symposium, "Cognitive/neuropsychological assessment is critical to learning disabilities—Or is it?" Reprinted with permission.

crucial. However, for adolescents and adults, in whom early intervention may have provided the maximum benefit or in whom intervention may have less impact on a well established and poorly treated learning disability, accommodations become very important.

## Accommodations

The most common accommodation is extended time on tests, typically from 25% to 100% of the standard amount of time, used to mitigate the impact of a learning disability on speed when reading, writing, or doing math. This becomes increasingly important, as tests become more complex in late elementary and middle school. Other common accommodations include the use of a computer for tests that require writing, extra breaks during lengthy tests, note-taking supports (permission to record lectures, access to PowerPoint presentations ahead of class, in-class note-taker), and course substitutions.

## Documentation Guidelines for High-Stakes Testing

Documentation guidelines have been published to guide neuropsychological assessment of adolescents and adults in secondary and postsecondary education. Originally developed by the Association on Higher Education and Disability and subsequently modified and revised by Educational Testing Service, testing agencies, such as the College Board, Law School Admission Council, Association of American Medical Colleges, and the National Board of Medical Examiners, as well as universities, have adopted these guidelines. Because requirements differ, guidelines should be consulted before completing an assessment. Also, because of new guidance for documentation issued by the Association on Higher Education and Disability, requirements for updated neuropsychological evaluations for secondary and postsecondary education students may be eased.

## Relevant Definitions

**Double-deficit model** A model of reading disabilities that hypothesizes two “single-deficit” subtypes of reading disabilities (phonological awareness or rapid automatized naming) with more limited reading impairments and one “double-deficit” subtype with more pervasive and severe impairments.

**Dysgraphia** Impairment in legible automatic letter writing, orthographic coding (i.e., storing written words and processing letters in them), and finger sequencing.

**Late-emerging reading disabilities** A pattern observed in approximately 40% of all students with reading disabilities in which problems are not observed until at least third grade and are often associated with the transition from “learning to read” to “reading to learn.” This pattern is sometimes referred to as the “fourth-grade slump.”

**Neural signature of dyslexia** A pattern observed in functional magnetic resonance imaging (fMRI) studies of reading in which underactivation is seen in Wernicke’s area, the angular gyrus, and striate cortex, and concurrent overactivation is observed in the inferior frontal gyrus.

**Response to intervention (RTI)** A model for defining learning disabilities based on the premise that students should receive adequate instruction with appropriate monitoring before being considered to have a learning disability.

**Visual word form area** A region in the left occipitotemporal cortex, including the middle part of the left fusiform gyrus, devoted to rapid processing of written words.

# Learning Disabilities Questions

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**NOTE:** Questions 14, 18, and 93 on the Full-Length Practice Examination are from this chapter.

1. The most prevalent learning disability is \_\_\_\_\_.
  - (a) mathematics disability
  - (b) reading disability
  - (c) written language disability
  - (d) nonverbal learning disability
2. You are seeing a child for an evaluation of dyslexia. The most important areas for assessment are phonological awareness, decoding, and \_\_\_\_\_.
  - (a) single-word reading, and spelling
  - (b) visual scanning, and spelling
  - (c) reading comprehension, and spelling
  - (d) single-word reading, and reading letters in their proper orientation and order
3. You are evaluating an employed adult who believes that he has a learning disability. Obtaining a detailed history from this adult \_\_\_\_\_.
  - (a) is crucial for establishing the developmental onset of these disorders, in combination with information on current functioning
  - (b) is desirable, although a definitive diagnosis may be made on the basis of current functioning
  - (c) should be supplemented with academic records whenever possible and combined with information on current functioning
  - (d) is not helpful without looking at records and obtaining a current evaluation of the adult's skills
4. Evidence-based intervention for learning disabilities is best established for \_\_\_\_\_.
  - (a) written language disabilities
  - (b) nonverbal learning disabilities
  - (c) mathematics disabilities
  - (d) reading disabilities
5. When evaluating a child for a learning disability who has not shown progress with standard instruction, which model has the best research support for diagnosis?
  - (a) aptitude-achievement discrepancy model
  - (b) intra-individual differences model
  - (c) low-achievement model
  - (d) response to intervention model
6. You are evaluating a college student who wants to apply for accommodations on the Law School Admission Test. He was diagnosed with a reading disability in second grade, had early intervention, and has had accommodations throughout his schooling. You want to be sure that your assessment conforms to the documentation guidelines. On which website should you look for the most current guidelines?
  - (a) National Joint Committee on Learning Disabilities
  - (b) Association on Higher Education and Disability
  - (c) Educational Testing Service
  - (d) Law School Admission Council

7. Although some researchers and clinicians believe there is usefulness in diagnosing \_\_\_\_, there is disagreement over whether it exists as a separate learning disability.
  - (a) nonverbal learning disability
  - (b) written language disability
  - (c) reading disability
  - (d) mathematics disability
8. A college student with a history of a written language disorder and accommodations for that is asking for use of a computer on writing tests. You would like to do a streamlined evaluation to document his need for accommodation. What skills will be most important to assess?
  - (a) visuospatial skills, organizational abilities, and spoken language comprehension
  - (b) handwriting speed, visuospatial skills, and visual learning and memory
  - (c) spelling, handwriting quality and speed, and vocabulary
  - (d) visuospatial skills, spoken language skills, and attention
9. You are seeing a college student who is having difficulty with pre-calculus in college. This is the first time that he has struggled in math. He has never been diagnosed with a mathematics disability and earned As and Bs in math through high school. The most important question to answer when taking a history is whether \_\_\_\_\_.
  - (a) he has had proper instruction in math and has built a foundation for his current college class
  - (b) the new demands of the math class are causing undue anxiety
  - (c) he might need to drop out and take a prerequisite for this class
  - (d) he is experiencing another source of stress in his life that could be causing difficulty in this class
10. You have evaluated an 18-year-old college freshman who has never been diagnosed with a learning disability and has never had formal academic support. However, review of her history, academic records, and standardized test scores is consistent with a reading disability. Which of the following would be the most appropriate recommendations for this student?
  - (a) work with a reading specialist and 50% additional time on tests
  - (b) vision therapy and the opportunity to take exams orally
  - (c) fifty percent additional time on tests and additional breaks during the test
  - (d) Irlen lenses to use when reading her textbooks and 100% extended time on tests
11. The double-deficit model of reading posits that deficits in which two skills are associated with persistent and severe reading disabilities?
  - (a) pseudoword decoding and reading of real words
  - (b) phonological awareness and verbal working memory
  - (c) rapid automatized naming and phonological awareness
  - (d) single word reading and reading comprehension

## Learning Disabilities Answers

1. **B—reading disability** *There is considerable research showing that reading disabilities are more prevalent than other learning disabilities, although some have suggested that mathematics disabilities may be almost as frequent.*

**References:** Fletcher, J. M., Lyon, G. R., Fuchs, L. S., & Barnes, M. A. (2007). *Learning disabilities: From identification to intervention*. New York: Guilford Press.

Fuchs, L. S., Fuchs, D., & Prentice, K. (2004). Responsiveness to mathematical problem-solving instruction among students with risk for mathematics disability, with and without risk for reading disability. *Journal of Learning Disabilities*, 27, 273–306.

National Center for Education Statistics. (2003). *National Assessment of Educational Progress, 2003, reading assessments*. Washington, DC: U.S. Department of Education, Institute of Education Sciences.

U.S. Department of Education, Institute of Education Sciences, National Center for Education Statistics. (2011). *National Assessment of Educational Progress—NAEP, 1990-2011 Mathematics Assessments*. Washington, DC: Author.

2. **A—single-word reading, and spelling** *Dyslexia is a specific reading disorder characterized by deficits in phonological awareness, fluent decoding and word reading, and spelling. It has nothing to do with visual processing or letter writing. Reading comprehension is affected, but the noted core deficits are key to assessment.*

**Reference:** Fletcher, Lyon, Fuchs, & Barnes (2007). Please refer to the reference for Answer 1.

3. **C—should be supplemented with academic records whenever possible and combined with information on current functioning** *A learning disability is a developmental disorder. It does not appear in adult life without evidence of earlier problems. The history can be the most important part of the assessment. However, memory is fallible, and so a better option is to review academic records. It is also necessary to show the expected academic and neuropsychological profile at the time of assessment. Adults who have no history of early learning problems but are reporting difficulty with reading, writing, or math most likely have another cause for their problems.*

**References:** American Psychiatric Association (2013). *Diagnostic and statistical manual for mental disorders*, 5th Edition (DSM-5). Washington, DC: American Psychiatric Press.

Mapou, R. L. (2009). *Adult learning disabilities and ADHD: Research informed assessment*. New York: Oxford University Press.

4. **D—reading disabilities** *Substantial research demonstrates the effectiveness of intervention to improve reading in children (and adults) with reading disabilities. There is far less research on interventions to improve written expression and math skills.*

**References:** Fletcher, Lyon, Fuchs, & Barnes (2007). Please refer to the reference for Answer 1.

National Joint Committee on Learning Disabilities (NJCLD). (2011). *Learning disabilities: Implications for policy regarding research and practice*. <http://www.ldonline.org/about/partners/njclcd#reports>

5. **C—Low-achievement model** *Of these four models, the low-achievement model has the best validity, especially when standard instruction has not been effective.*

**References:** Fletcher, J. M. (2009). Dyslexia: The evolution of a scientific concept. *Journal of the International Neuropsychological Society*, 15, 501–508.

Fletcher, Lyon, Fuchs, & Barnes, 2007. Please refer to the reference for Answer 1.

6. **D—Law School Admission Council** *Guidelines change periodically. The best resource is always the website for the organization that administers the standardized test.*

**Reference:** Mapou (2009). Please refer to the reference for Answer 3.

7. **A—Nonverbal learning disability** *There is far less empirical support for nonverbal learning disability as a separate learning disability. The educational impact is less specific, and there is overlap between the features of nonverbal learning disability and ADHD.*

**References:** Fletcher, Lyon, Fuchs, & Barnes (2007). Please refer to the reference for Answer 1.

Pennington, B. F. (2009). *Diagnosing learning disorders: A neuropsychological framework* (2nd ed.). New York: Guilford Press.

Spreen, O. (2011). Nonverbal learning disabilities: A critical review. *Child Neuropsychology*, 17, 418–443.

8. **C—Spelling, handwriting quality and speed, and vocabulary** *Research on college students with dyslexia has shown that these four skills are highly predictive of writing problems. Although individuals with poor visuospatial skills, organizational skills, and attention can have difficulty with writing, these are less important components of an assessment.*

**References:** Connelly, V., Campbell, S., MacLean, M., & Barnes, J. (2006). Contribution of lower order skills to the written composition of college students. *Developmental Neuropsychology*, 29, 175–196.  
 Gregg, N., Coleman, C., Davis, M., & Chalk, J. C. (2007). Timed essay writing: Implications for high-stakes tests. *Journal of Learning Disabilities*, 40, 306–318.

9. **A—He has had proper instruction in math and has built a foundation for his current college class** *This student is unlikely to have a learning disability, if he made it to college with As and Bs in math. Although the other areas should certainly be addressed, the answer most relevant to a specific problem in math is A.*

**Reference:** Fletcher, Lyon, Fuchs, & Barnes (2007). Please refer to the reference for Answer 1.

10. **A—Work with a reading specialist and 50% additional time on tests** *Given that this is a first-time diagnosis and assuming the problems are not severe, these would be the best recommendations. Recommendations should always include intervention and not just accommodations, so that the student can improve her skills. Vision therapy and Irlen lenses have been shown to not be effective. Individual testing and additional breaks are more appropriate accommodations for students with attentional disorders.*

**References:** Gregg, Coleman, Davis, & Chalk (2007). Please refer to the reference for Answer 8.  
 Handler, S. M., Fierson, W. M., & the Section on Ophthalmology and Council on Children with Disabilities. (2011). Learning disabilities, dyslexia, and vision. *Pediatrics*, 127, e818–e856.  
 Mapou (2009). Please refer to the reference for Answer 3.

11. **C—Rapid automatized naming and phonological awareness** *The presence of deficits in both phonological awareness (single deficit) and rapid automatized naming (single-deficit) contribute to the double-deficit condition, which is associated with persistent reading disability.*

**Reference:** Wolf, M., & Bowers, P. G. (1999). The double-deficit hypothesis for the developmental dyslexias. *Journal of Educational Psychology*, 91, 415–438.

## Recommended Readings

- Fletcher, J. M., Lyon, G. R., Fuchs, L. S., & Barnes, M. A. (2007). *Learning disabilities: From identification to intervention*. New York: Guilford Press. [Primarily chapters 4 through 9]  
 Mapou, R. L. (2009). *Adult learning disabilities and ADHD: Research-informed assessment*. New York: Oxford University Press.  
 Pennington, B. F. (2009). *Diagnosing learning disorders: A neuropsychological framework* (2nd ed.). New York: Guilford Press. [Especially chapters 6, 12, 14, and 15]  
 Swanson, H. L., Harris, K. R., & Graham, S. (Eds.). (2013). *Handbook of learning disabilities* (2nd Ed.). New York: Guilford Press.  
 Wolf, L. E., Schreiber, H., & Wasserstein, J. (Eds.). (2008). *Adult learning disorders: Contemporary issues*. New York: Psychology Press.